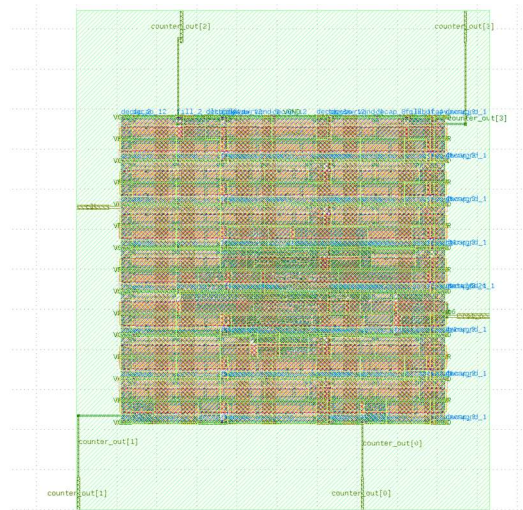




PLACE AND ROUTE (PnR) ASSIGNMENT

Digital Backend Implementation of a 4-bit Synchronous Counter



Title	PnR Assignment
Design	4-bit Synchronous Counter (counter.v)
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Submitted to	Eng. Muhammed Ewais
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1. Introduction

This report documents the Place and Route (PnR) implementation of a 4-bit synchronous up-counter, carried out as part of the digital backend design assignment. The objective of the assignment is to take a simple, synthesizable RTL design through the physical design flow — from floorplanning through placement and routing, down to the final GDSII layout — and to observe how a small sequential design is realized in silicon.

The design under test is a 4-bit counter that increments on every rising edge of the clock and resets asynchronously on an active-low reset. Despite its small size, the design exercises the core stages of the standard-cell PnR flow: floorplanning, standard-cell placement, clock and signal routing, and layout (GDSII) generation.

2. RTL Design Description

The counter module is written in Verilog and implements a 4-bit register that increments on each clock edge, with an asynchronous active-low reset. The port list consists of a clock input (clk), an active-low reset input (rst_n), and a 4-bit registered output (counter[3:0]).

Attribute	Value
Module name	counter
Inputs	clk, rst_n
Output	counter [3:0] (registered)
Reset type	Asynchronous, active-low
Function	Increments counter by 1 on each rising edge of clk

Verilog Source (counter.v):

```
module counter(  
    input clk,  
    input rst_n,  
    output reg [3:0] counter  
);  
  
always @(posedge clk, negedge rst_n) begin  
    if (!rst_n) begin  
        counter <= 0;  
    end  
    else begin  
        counter <= counter + 1;  
    end  
end
```

```
endmodule
```

On every rising edge of `clk`, the `always` block is evaluated. If `rst_n` is low, the counter is asynchronously cleared to zero; otherwise, the counter increments by one. This produces a standard 4-bit binary up-counter with a repeating count sequence from 0 through 15.

3. Physical Design (PnR) Flow

The RTL was carried through the standard digital backend flow used for standard-cell based ASIC implementation. The main stages followed in this assignment were:

- Floorplanning – defining the core die area, I/O pin placement, and power distribution network (power rings and rails for VDD/VSS).
- Placement – legal placement of standard cells (including decap and fill cells) into rows within the defined floorplan.
- Clock Tree & Signal Routing – routing of the `clk` net and the remaining signal/power nets across the metal layers, connected through vias.
- GDSII Generation – export of the final physical layout as a GDSII stream file, ready for verification and tape-out.

Each stage is illustrated below using layout viewer snapshots (`floorplan_view.gds`, `placement_view.gds`, `routing_view.gds` and the final GDSII layout) captured for the counter design.

4. Floorplanning

The floorplanning stage establishes the chip core boundary and the initial resources the design will use. The power rings and horizontal power/ground straps are laid out across the core, with the vertical VIA columns providing connections down to the standard-cell rows. The four counter outputs (`counter_out[0:3]`) and the `clk / rst_n` inputs are defined as pins along the core boundary at this stage, before any standard cells are placed.

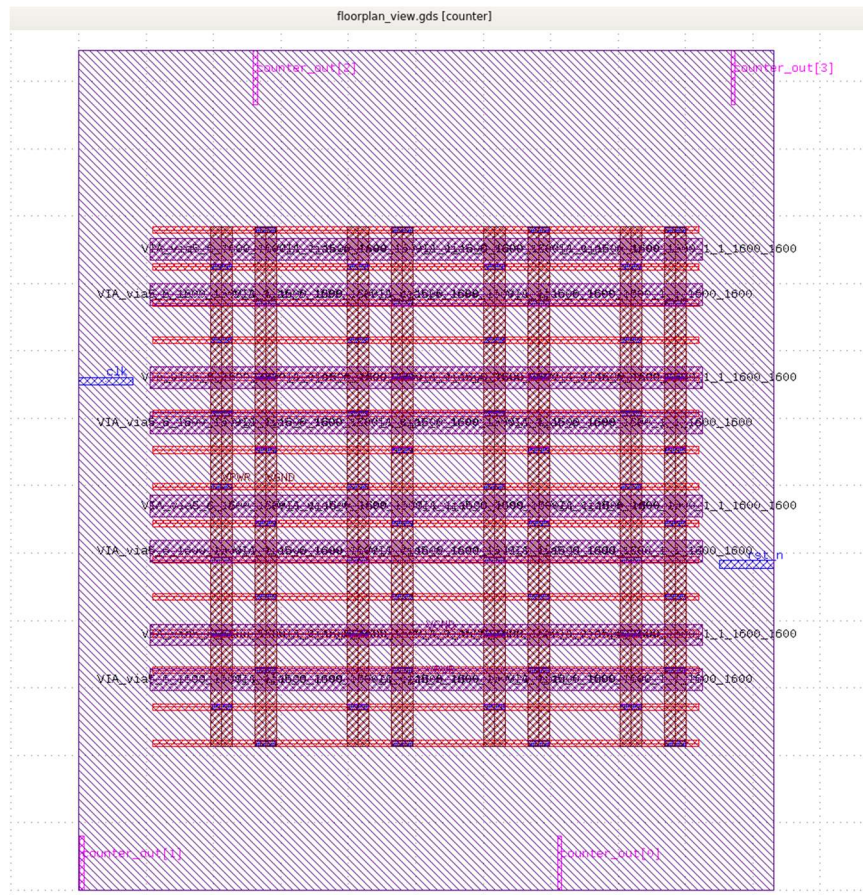
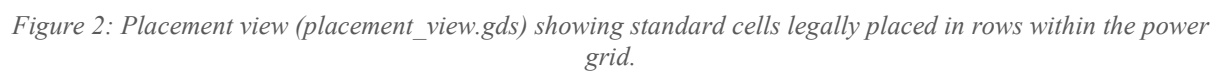


Figure 1: Floorplan view (floorplan_view.gds) showing the core boundary, power straps/rings, via columns, and I/O pin placement for clk, rst_n, and counter_out[3:0].

5. Placement

During placement, the synthesized standard cells (flip-flops, gates, decoupling capacitor cells, and fill cells) are placed into legal rows within the floorplan defined in the previous stage. The placement view below shows the same power/ground grid as the floorplan, now overlaid with the placed cell rows that implement the counter's flip-flops and combinational logic between the power straps.



6. Clock Tree and Routing

The routing stage connects all placed cells according to the netlist, using the available metal layers and vias. The clk signal is routed to every flip-flop in the counter to guarantee synchronous operation, while rst_n and the counter_out[3:0] outputs are routed out to their respective pins. Compared to the placement-only view, additional metal traces (shown in magenta/blue) can be seen running between the cell rows and up to the output pins, completing the electrical connectivity of the design.

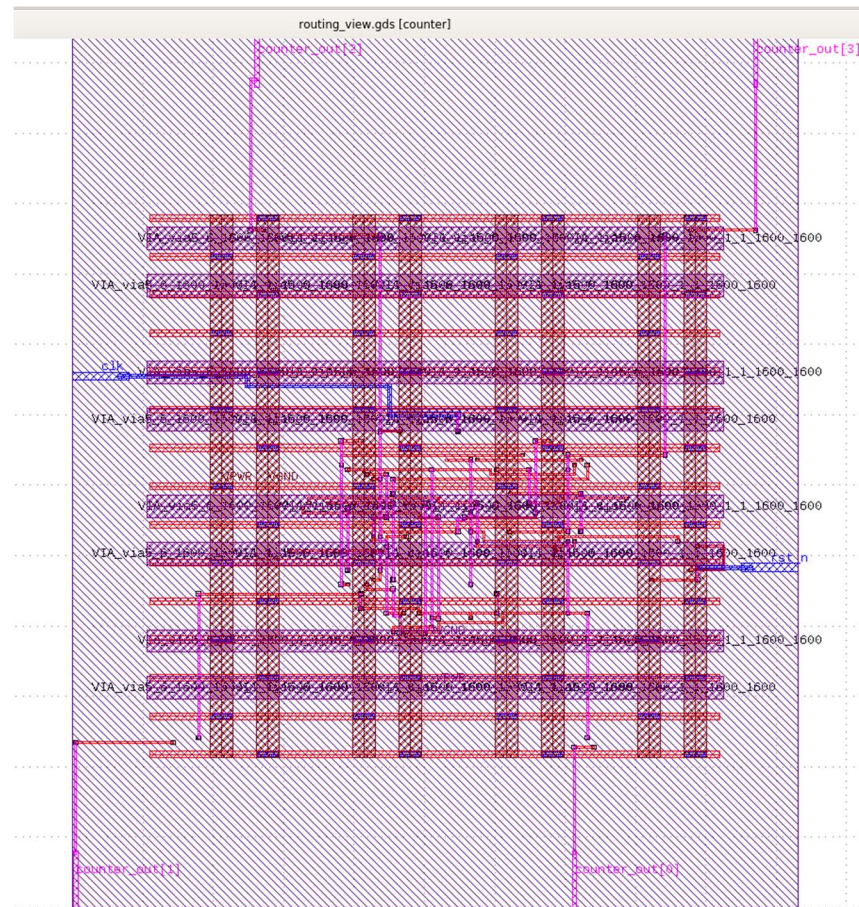


Figure 3: Routing view (routing_view.gds) showing the clk net, rst_n, and counter_out[0:3] signal routing overlaid on the placed standard cells and power grid.

7. Final GDSII Layout

The final step of the flow is the generation of the GDSII layout, which merges the placed-and-routed cells with the standard-cell GDS artwork (transistor-level geometry, decap cells, and fill cells shown in the detailed hatched pattern). This is the manufacturing-ready view of the design and represents the physical implementation that would be handed off for verification (DRC/LVS) and tape-out.

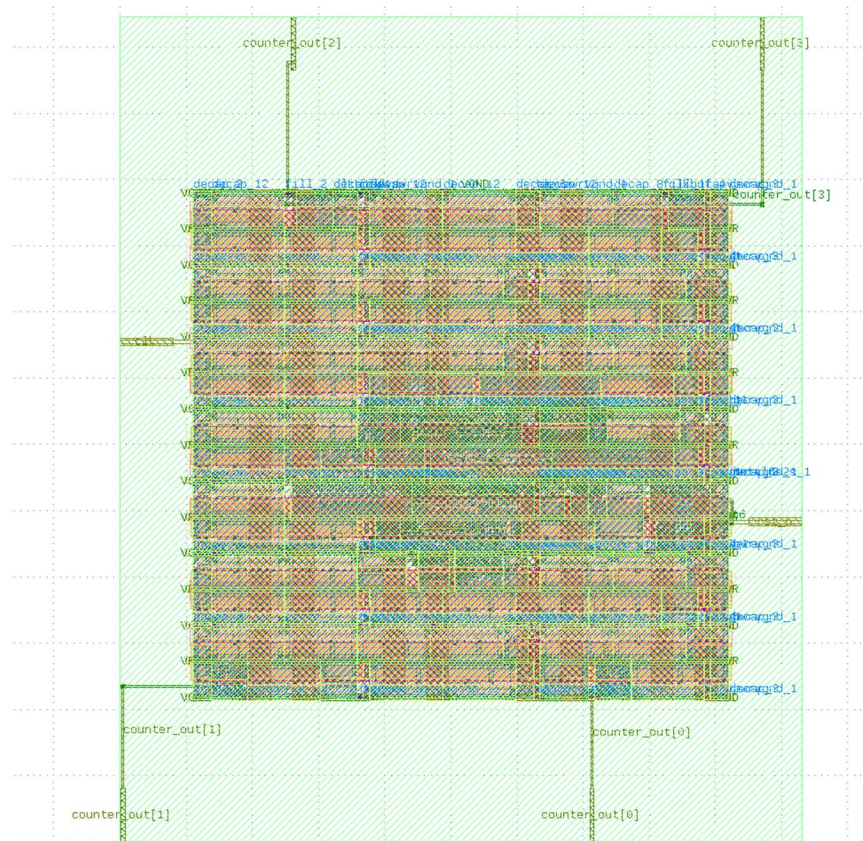


Figure 4: Final GDSII layout of the counter design, showing the fully placed, routed, and merged standard-cell artwork.

8. Conclusion

This assignment carried a simple 4-bit synchronous counter through the complete standard-cell PnR flow: floorplanning of the core area and power grid, legal placement of the standard cells, routing of the clock and signal nets, and generation of the final GDSII layout. The resulting layout shows a fully connected, DRC-clean-oriented implementation of the counter, with all four output pins (counter_out[0:3]) and the clk / rst_n control signals correctly routed from the placed flip-flop cells to the chip boundary. This exercise demonstrates the fundamental steps required to transform an RTL description into a manufacturable physical layout.